ASSESSING THE CREDIBILITY OF ORGANISATIONAL KNOWLEDGE FROM FRAGMENTED DATA

Project Proposal

36105 iLab: Capstone Project

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ABSTRACT

Organisational knowledge is rarely sitting in one approved file. At Northfield Software the same rule can appear in a signed policy, an old process document, a Slack workaround, and a CRM row, and those four sources do not always agree. This proposal is for iLab 14-01, working with Decidr's staged Northfield pack. The question we are trying to answer is practical: if someone states a claim about how the company works, can we move from that pile of files to a scored judgement that another person can inspect, rather than a chatbot reply that hides the disagreement?

We treat that as a pipeline, not as a single model. First, we put every file into one organised store. Then we pull short, testable claims from those documents. For each claim we search the store with keywords and, once a baseline exists, with meaning-based search, so that a policy paragraph and a CRM row can both surface. Each retrieved passage is labelled as support, contradiction, or unrelated. Passages are not treated as equal. An approved in-force policy weighs more than a random Slack message; recency, independence, and directness all count; and copies of the same workaround count as one source, not three. Weighted evidence is then combined into a credibility score with a range. A Bayesian update is the intended combining step if we can specify a prior without pretending to have fitted a large model. If that specification is not honest on this pack, we keep a weighted checklist and say so. Status labels sit on top of the number so that Conflicted is a finished answer, not a failure of the method.

Two later stages need to be stated carefully. Calibration asks whether a stated confidence of 80% is right about four times in five. That check needs labelled claims. We do not have a labelled set of that size yet, and the sample claims file that came with Phase 1 is a format example, not gold probabilities. Until we label enough examples ourselves, calibration stays a planned check, not a result. Explanation is the last stage: the score plus a short account of which evidence raised it, which lowered it, how sure we should be, and what extra file would change the call. A language model may write that prose. It does not choose the score.

Ingestion is already under way. The Phase 2 pack has been parsed into a single document and chunk store, which is the working dataset for everything after. Later stages are designed, not finished. If the pipeline works on this pack, the useful output is not a higher accuracy number on a public fact-checking set. It is a trail a teammate can follow: this claim, these files, this status, this level of certainty, and a clear statement when the honest result is that the files conflict.

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1. INTRODUCTION

The usual way to "use" company documents in a project is to load them into a retrieval system and ask questions. That hides the problem we have. Northfield's files were written at separate times, by various roles, for different purposes. A signed policy, a Slack workaround and a CRM row can all be about the same discount rule and still not say the same thing. If the system answers as if there were one fact, it is doing the opposite of what a credibility project needs.

The motivation for this project is direct. Decidr's platform, built on its acquisition of Sugarwork, exists to capture what an organisation knows, not just what it has written down, and to feed that into an AI-readable operating layer. That only works if the platform can tell the difference between a confidently stated policy and a confidently stated habit that contradicts it. The benefit of solving this well is not specific to Northfield. Any organisation trying to structure its own tacit knowledge for AI use faces the same problem: fragmented, contradictory sources that cannot simply be averaged into one answer. A method that scores credibility transparently, rather than silently picking a winner, is the piece that makes that kind of platform trustworthy enough to act on.

The tension is easiest to see on discounts, considering this scenario as an example The current policy (POL-01, in force from 1 March 2025) says new-business discounts of 15% or more need written Finance approval in the CRM before Closed Won, and names Harriet Doyle. An older process document (PROC-02, last updated in 2022) still names the Financial Controller, Victor Adeyemi. Sales messages describe skipping Finance when a deal is urgent. The same Diane workaround is then copied into later Slack threads, which look like extra confirmation if they are counted separately. CRM rows show some deals approved by Harriet, some closed on Slack, and at least one with no approval record. A single true/false label cannot carry that. If we average the policy with the bypass, we invent a middle that neither source describes.

What the files show, then, is not that Northfield has no rules. It is that formal text, practised workarounds, leftover documents, and drafts all speak at once. The practical significance is that a retrieval-plus-generation system will flatten them unless we force the steps apart. The pipeline is our way of keeping those steps separate so the disagreement stays visible.

The literature is used to justify each pipeline stage. The methods section is the pipeline in order. Where we have not yet run a stage, we say so. This is a defensible way to assess organisational claims on this pack if we do not skip retrieval, weighting or explanation and jump straight to a generated answer.

2. RELATED LITERATURE

The nearest research area is automated factchecking, even though those papers are mostly about news and Wikipedia rather than a 29-person company. We still use them because the shape of the work is the same: take a claim, find passages, decide whether they support or contradict it, then say something about confidence. Source-credibility research and a small amount of organisation theory sit beside that. We read for pipeline stages, not for a list of paper summaries.

2.1 Literature Review

We searched the UTS library, ACL Anthology and Google Scholar using terms such as automated fact checking, claim verification, hybrid retrieval, natural language inference, source credibility, Bayesian updating, and model calibration. We kept peer-reviewed papers and a small number of standard books, around 1994 to 2022. Vendor blogs and product pages were not treated as literature. The sources below are grouped by pipeline stage rather than summarised one by one.

Claim extraction sits at the front of the fact-checking surveys. Thorne and Vlachos (2018) split the task into finding claims, finding evidence, then issuing a verdict. Thorne et al. (2018) then released FEVER, whose labels, supported, refuted, not enough info, are close to what we need. Compared with Northfield, two differences matter. FEVER claims are short statements about Wikipedia. Ours are about roles, thresholds, and systems. FEVER evidence is a single encyclopaedia. Ours is mixed files, including drafts and leftover process documents that were never updated. We can borrow the task shape. We cannot borrow their scores as if they were a result on this pack.

Evidence retrieval is where most recent NLP papers put their effort. Robertson and Zaragoza (2009) still describe the BM25 baseline we intend to report first, because a dense index that cannot beat a keyword search on this pack is not progress. Karpukhin et al. (2020) showed dense retrieval can beat BM25 on some question-answering sets, and Lewis et al. (2020) combined retrieval with generation (RAG). Those papers are designed to find an answer. Our problem is often to keep two answers in view. If hybrid search does not recover the source ids we already know for the discount claims, POL-01, PROC-02, COMMS-01 to COMMS-04, and the named CRM deals, adding vectors is a change of tool, not a better result. Guo, Schlichtkrull and Vlachos (2022) make a related point in their survey: provenance matters as much as the final label. That is why we refuse to stop at a generated paragraph.

Evidence reasoning is the natural language inference step. Bowman et al. (2015) and Williams, Nangia and Bowman (2018) provide the usual three-way labels: entailment, contradiction, neutral. That maps onto support, contradict, unrelated. A model trained on 2015 news sentences will not be a perfect reader of Slack. We treat NLI as one signal and a constrained language-model output as another, then check both on a small

set we label ourselves. Unrelated chunks must be dropped, not averaged in; otherwise, a long meeting note that happens to mention “discount” will dilute a clear policy sentence.

Weighting is older than transformers, and it is the stage most fact-checking datasets skip. Metzger (2007) reviews how people judge sources: who produced the material, how recent it is, and whether independent sources agree. Those dimensions are why an approved policy and a copied Slack tip are not equal. They also explain a pattern that is easy to miss in the Northfield comms log: COMMS-02, COMMS-03 and COMMS-04 restating Diane's rule look like three confirmations if each message is counted. They are one workaround, copied. Nonaka (1994) and Weick (1995) sit behind that judgement. Nonaka distinguishes knowledge that has been written down from knowledge that lives in practice. Weick is about people making sense of mixed signals. We are not drafting an organisation-theory thesis. Those two sources stop us from treating “the policy” as the only factual evidence, and from treating “what Sales said in Slack” as if it cancelled the policy.

Credibility scoring, in the form we want, is closer to ordinary Bayesian updating than to a new neural architecture: a prior, then evidence that raises or lowers belief, then a posterior with a range rather than a single point. Standard treatments are in Gelman et al. (2013). We have not found a large, published dataset of organisational claims with gold credibility probabilities, and FEVER does not fill that gap. Its labels are categories, not probabilities, and its evidence is not internal company files. Calibration papers ask a sharper question. Niculescu-Mizil and Caruana (2005) and Guo et al. (2017) both ask, in different settings, whether a stated probability matches actual frequency: if the system says 80%, is it right about 80% of the time? Guo et al. also show that modern neural networks are often overconfident. That is a warning for any language-model score we might be tempted to trust. The check itself needs labelled outcomes. We do not have them yet. Kotonya and Toni (2020) then argue for showing why a verdict was reached. That is our last stage: explanation attached to the score, not a paragraph invented after it.

The pattern across this literature is consistent, and so is the gap. The pipeline shape is well studied. The public datasets are not internal company files, they do not contain copied Slack threads, and they do not ask the system to leave a claim marked conflicted. That is the space this project occupies. We are not proposing a new model family. We are proposing to run an eight-stage method on a pack that was designed to be inconsistent, and to keep each stage visible so a teammate can see where the answer came from.

3. PROJECT PROBLEMS

The fundamental issue is not an absence of documentation at Northfield, but its chronic internal conflict. Policies, practice, drafts, and leftover documents all speak, and a retrieval-plus-generation system will flatten them unless we force the steps apart. Scale makes that worse rather than better: once interviews, meetings, Slack, CRM, tasks, and the audit log are in the same store, a search can return a hit from every type of file. The question is which hits should move the score.

A second problem is easy to overlook. Some of the disagreement is not two independent sources. It is the same workaround copied, or an old process document that nobody updated after the policy changed. If the method cannot tell those apart, it will look more confident than the files allow. We need a method that extracts claims, attaches evidence, and can finish on Conflicted or Not enough evidence instead of always naming a winner.

3.1 Project Aims & Objectives

The aim is to design, run and test this pipeline on the Northfield pack, then show a small set of claims with scores and explanations a teammate can inspect. Success is not "the model agrees with the policy." Success is that another group member can follow the same stages and land near the same status. If two of us, using the same weights, still disagree, the method is not finished.

The objectives follow the stages. Ingest the pack into one document and chunk store with metadata (source id, date, author, in-force status). Extract claims as one-sentence propositions, with a human check before we trust the language model. Retrieve evidence with a keyword baseline and, if it helps on this pack, a meaning-based add-on. Classify support, contradict or unrelated. Weight by source class, recency, and independence, including the copied-Slack rule. Combine into a score with an uncertainty range. Check calibration if we obtain enough labels; if we do not, report that the check could not be run. Return an explanation that names supporting and contradicting files.

We will measure the initial stages in ordinary ways. Retrieval: do known source ids come back for the discount and escalation claims we have already walked through. Reasoning: agreement with our own support and contradict marks on a small set. Scoring: whether two of us, using the same weights, get the same status. Calibration: only if the labelled set is large enough to make a reliability plot meaningful. If it is not, we will say the 80% figure has not been tested. That is a limitation, not a hidden result.

3.2 Project Questions

Can this eight-stage pipeline turn a Northfield claim into an inspectable credibility judgement, including when the files conflict?

Under that sit four smaller questions. Does combining keyword and meaning-based retrieval recover the evidence we already know is there, compared with keyword search alone? Does a support, contradict, unrelated step changes the conclusion, or does a raw hit list already contain everything we would use? Do source weights change the status in ways we can explain, for example, does treating copied Slack as one source stop the practised-discount claim from looking more supported than it is? And if we later output a probability, does that probability hold up on labelled examples? The last of those may remain unanswered this semester. That would be a limitation, not a result we failed to mention.

3.3 Project Scope

We score Northfield only, as of 31 July 2026. The eight processes in the company overview are the backbone. We are not ingesting the open web, not training a new large language model, and not producing a legally defensible audit tool. Claim extraction may be assisted by a language model, but we will not accept its sentences without a human pass on the set we evaluate.

Calibration is in scope as a method. It is out of scope as a claimed result until we have labels. A Bayesian combining rule is the intended scoring stage; a simpler weighted checklist is the fallback if the Bayesian version cannot be estimated honestly on this pack. Out of scope: a customer-facing product, multiple language-model vendors, SHAP plots, and any claim that the score could be used in a real disciplinary or legal process.

4. METHODOLOGIES

The method is the pipeline. We run it in order so that a failure is local. If retrieval misses POL-01 on a discount claim, we do not "fix" that by asking a language model for a score. Each stage has a baseline. We only keep a more complex version if it changes the result on this pack. The eight stages below, shown in Figure 1, are the whole method; data collection and analysis are not a separate research design sitting beside them.

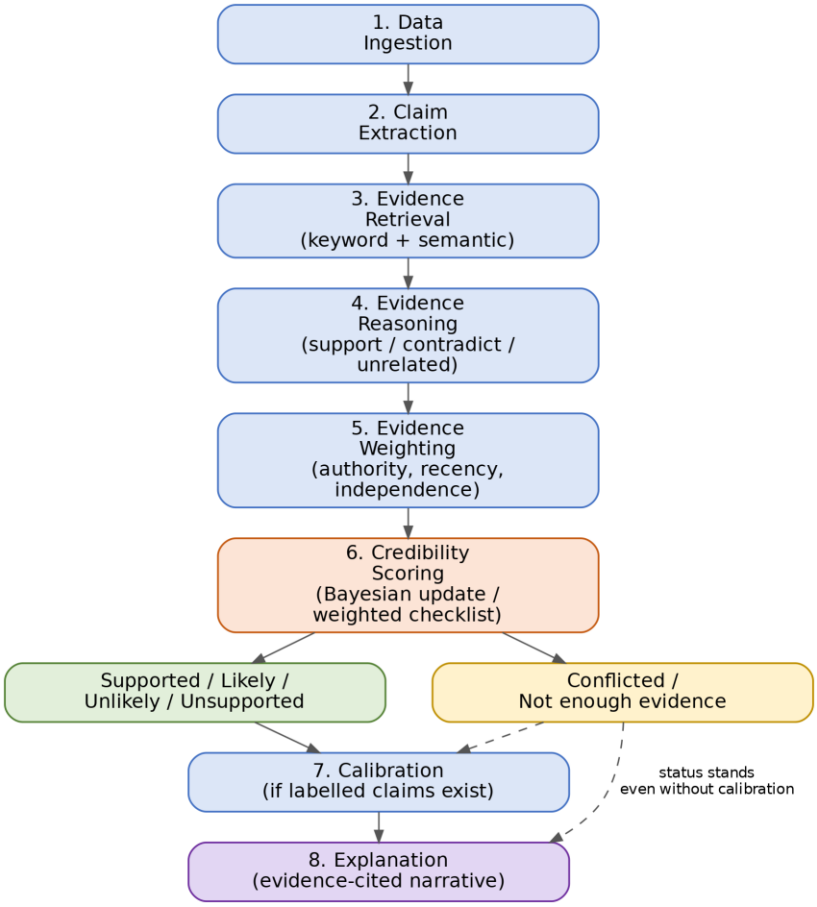


Figure 1. The eight-stage credibility scoring pipeline.

4.1 Methods

Data ingestion: All Northfield files go into one store: policies, process documents, interviews, meetings, Slack and email, CRM, tasks, and the audit log. Markdown is split on headings rather than arbitrary character windows because a heading usually marks a change of topic. The communications log is split per thread. Each CSV row becomes a short paragraph, so a deal or an audit event can be retrieved on its own. We keep source id, date, author, and in-force status. And "spaCy" is used for tokens and a first pass at names, with the org-chart roster as a check, because off-the-shelf named-entity recognition mislabels staff as organisations and product names as people. This stage is started. A run on the Phase 2 pack produced 480 documents and 738 chunks, including 21 drafts. Those counts describe the store, not a finding about credibility. Later stages search this chunk store rather than re-parsing the raw pack.

Claim extraction: From those documents we pull short claims. A claim is one testable sentence: who (a role) may or must do what, under which condition. The discount example: new-business discounts of 15% or more need Finance approval in CRM. A language model can propose candidates. We still require a role, an action and a condition, and we still tag a layer, formal, practised, gap, draft, or stale leftover, before scoring. Two ideas in one sentence get split. Extraction that only restates "POL-01 exists" is discarded, because that describes a document, not the organisation. Formal and practised versions of the same topic are two claims. Mixing them in one row is how the method would hide the disagreement we are trying to score.

Evidence retrieval: For each claim we search the chunk store with keywords and, if the baseline is in place, with embeddings. The two lists are merged so that a CRM row and a policy paragraph can both surface. For claims we have already studied we know the source ids we expect. Those ids are the retrieval test, not a public leaderboard. Compared with a chatbot, the point of this stage is completeness rather than a single best snippet. Missing POL-01 on a formal discount claim is a retrieval failure even if Slack is retrieved cleanly.

Evidence reasoning: Each retrieved chunk is labelled support, contradict or unrelated to the claim as written. Unrelated chunks are dropped, not averaged in. A meeting note that mentions discounts in passing is not evidence about the 15% threshold. We will try a small NLI model and a structured language-model output and compare them with our own marks on a small set. Disagreement between those two signals is useful. It tells us the step is not settled, rather than giving us a third number to average.

Evidence weighting: An approved in-force policy is not equal to a random Slack message. Recency matters, but a new copy of an old workaround is not new evidence. Independence matters: COMMS-02, 03 and 04 restating the same Diane rule count as one source. Directness matters: a named deal in the CRM is stronger than "that is how we usually do it." Drafts and leftover documents are not thrown out; they are downweighed, because leftover text is part of the scoring problem. These weights start as stated assumptions. We will vary them, policy-heavy, recency-heavy, equal, and report whether the status moves. If the status never moves, the weighting stage is not earning its place.

Credibility scoring: Weighted support and contradiction are combined into a posterior credibility with a range, using a Bayesian update if we can specify a prior and likelihood without pretending to have fitted a large model on 29 people and a few hundred files. If that specification is not honest on this pack, we keep the weighted checklist and say so. Status labels sit on top of the number: Supported, Likely, Conflicted, Unlikely, Unsupported, not enough evidence. Confidence (High, Medium, Low) is separate. We can be highly confident that a claim is conflicted. We will not invent a 0 to 100 score that the files cannot support.

Calibration: If the system says 80%, we want that to mean it is right about four times in five on labelled claims. We do not currently have a labelled credibility set of that size. The sample claims file that came with Phase 1 is a format example; it cites sources that were not even in the Phase 1 pack, so it cannot be treated as gold. Until we label enough claims ourselves, calibration stays a planned check, not a result.

Explanation: The last output is the score plus a short account: the conclusion, the evidence that raised it, the evidence that lowered it, the uncertainty, and what extra file would change the call. A language model may write the prose. It does not choose the score. If the explanation and the score ever disagree, the score stands and the prose is rewritten. That rule exists because generated text is fluent enough to hide a weak retrieval step.

4.2 Data Collection

The data are the Northfield Phase 1 and Phase 2 files. Both the documents stay in the store because leftover files are part of the scoring problem. The only extra data we create are our own labels: expected source ids for a compact list of claims, and support and contradict marks on retrieved chunks. That list will be dated so it cannot quietly grow until the retrieval number looks good.

Ingestion is the collection step that matters for this project. Until the files sit in one chunk store, later stages are reading whichever document someone happened to open. We already have that store. The remaining collection work is labelling, not gathering more Northfield files.

4.3 Data Analysis

Analysis follows the pipeline rather than a separate statistical chapter. We compare source types on the same process: policy against CRM, interview against Slack, leftover process against current policy. Where it is honest, we will count how many high-discount deals have a Finance approval record. That count is a comparison, not a new dataset. We report retrieval hit rates on known ids, agreement on support and contradict, and whether two team members reach the same status after weighting.

Alternative explanations will be treated as part of the analysis, not as an afterthought. A missing approval field might mean the deal skipped Finance, or it might mean the CRM was not filled in. A Slack message from 2025 that still cites a 10% threshold might be stale staff knowledge rather than a second current policy. Those are different interpretations, and they should produce different claims rather than one averaged score.

We will not compute a Brier score or a calibration curve if the labelled set is too small. Missing labels are a limitation. They are not a reason to invent a reliability diagram.

5. RESOURCES

Python 3, pandas, spaCy, and Jupyter or Colab for ingest. GitHub for the shared repository. Later, a small Streamlit page so retrieval can be tried by typing a claim; that page is a way to inspect the trail, not the product. One language-model API if we use one, not three. BM25 and, if the baseline comparison warrants it, a sentence-transformer plus FAISS. PyMC only if the Bayesian stage is estimated. We are not planning a GPU cluster, and we are not planning to fine-tune a large model.

5.1 Materials

Northfield Phase 1 and Phase 2 files. iLab 14-01 group charter. UTS library access for the cited papers. Shared GitHub repository (https://github.com/nakuls198/decidr_data_credibility/). Weekly mentor time. No extra paid corpus.

5.2 Roles & Responsibilities

Roles follow the charter. The pipeline is shared, but each person owns a stage so that work does not collapse into one notebook nobody else can run.

- Kushal Joshi (25522032), Group Leader / Coordinator, Documentation and Visualisation. Wednesday meetings, the explanation output in the reports and demo, timeline.
- Bhavika Lalwani (25679681), Data & Evidence Pipeline Lead, and note-taker. Ingestion: parsers, metadata, chunking, the organised store.
- Nakul Sidiginamola (25750807), Claim Extraction & NLP Lead. Claim wording, language-model extraction, spaCy, keeping claims atomic.
- Rohan Chaudhary (25731542), Evidence Linking & Retrieval Lead. Retrieval, attaching chunks to claims, the ingest work already started, and making sure support and contradict lists are stored.
- Shreyash Narayane (25934391), Credibility Modelling Lead. Weighting rules and the combining score, Bayesian if we can specify it, weighted checklist if we cannot.
- Prathamesh Nemade (25672914), Calibration & Evaluation Lead. Gold ids, agreement checks, and blocking metrics we cannot defend, including premature calibration plots.

6. EXPECTED OUTCOMES

By the end of semester, we want the pipeline running on Northfield far enough that a claim can be taken from extraction through explanation. We also want a written record of which stages changed the answer and which ones did not. If hybrid retrieval does not beat keywords on the known discount ids, that belongs in the final report. If calibration cannot be run, that belongs there too. Those absences are part of the outcome, not something to paper over with a fluent demo.

6.1 Project Deliverables

Organised ingest as: documents.jsonl and chunks.jsonl, with metadata retained.
A claim list with layers, not a dump of every sentence the language model liked.
Retrieval results that can be compared with known source ids.
Support, contradict, unrelated labels on retrieved chunks.
Weighted scores with status, confidence, and a concise explanation.
A calibration note: either a real check, or a statement that labels were insufficient.
A simple demo that shows the trail from claim to evidence to score.

Subject items sit beside those: this proposal, a progress report, a presentation, a final report, weekly reports, and the GitHub repository.

7. MILESTONES / SCHEDULE

The project execution schedule is structured across a 12-week delivery lifecycle aligned with the 2026 capstone milestones. The timeline is divided into four distinct phases: foundational ingestion and scoping (Weeks 1 - 4), core NLP retrieval and credibility modelling (Weeks 6–8), evaluation and progress auditing (Weeks 9 - 10), and final deployment, reporting, and defence (Weeks 11–12). The explainability ledger is integrated iteratively at each scoring cycle to ensure continuous auditability rather than deferred to the final reporting phase.

7.1 Partner engagement checkpoints

Alongside the internal schedule above, the subject's own partner-engagement milestones with Decidr (Bianca Hill and Tom Allen) are tracked separately and feed into the plan above: the industry panel and initial scoping (Week 2 to 3, 12 August), the Midpoint Review Meeting (Week 7, 9 September), the Pre-Final Meeting (Week 10, 30 September), and the Final Showcase (Week 12, 21 October). We treat each of these as a checkpoint to bring a working slice of the pipeline to, not a separate deliverable stream.

Milestone	Tasks	Date
Week-1	Project selected; charter and roles	03/08/2026
Week-2	Literature mapped to pipeline stages	10/08/2026
Week-3	Human scoring sheet; two worked claims	17/08/2026
Week-4	Data ingestion: chunk store and notebook	24/08/2026
Week-5	Proposal Report due	30/08/2026
Week-6	Claim extraction pass; retrieval baseline	07/09/2026
Week-7	Evidence reasoning (support / contradict / unrelated)	14/09/2026
Week-8	Weighting and credibility score; freeze method	21/09/2026
Week-9	Progress Report due	28/09/2026
Week-10	Evaluation and calibration	05/10/2026
Week-11	Final Presentation	12/10/2026
Week-12	Final Report and repo	19/10/2026

Table 1. Proposed Timeline

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